WORKED SOLUTION

STEP III 2019, Question 12

Probability and Statistics • Random Subsets

STEP III 2019, Question 12

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 9.12, *Random Subsets*: https://beyondstep.keithmcnulty.org/ch09/random-subsets.html

The key idea

Choosing a subset of S uniformly at random is the same as going through the integers $1, 2, \ldots, n$ one at a time and tossing a fair coin for each to decide whether it goes in. Once we see this, every event in the question becomes a statement about each integer separately, the integers behave independently, and the answer is a single small probability raised to the power n.

Solution

Counting the subsets

A subset of S is fixed completely by deciding, for each of the n integers, whether it is in or out. There are 2 choices for each integer, made independently, and different patterns of choices give different subsets. So the number of subsets, counting $\varnothing$ (everything out) and S (everything in), is

$$2 \times 2 \times \cdots \times 2 = 2^n.$$

? **Ask yourself** Why may we treat the integers as independent coin tosses?

Let A be uniform on T. Each of the 2^n in/out patterns is one subset, so each pattern has probability $2^{-n} = (\frac{1}{2})^n$. That is exactly the probability of a given pattern when we toss n independent fair coins. So the events "$1 \in A$", "$2 \in A$", ..., "$n \in A$" are independent, each with probability $\frac{1}{2}$. Since $A_1, \ldots, A_m$ are chosen independently of each other too, the mn events "$i \in A_k$" are all independent, each with probability $\frac{1}{2}$.

Part (i)

Exactly half of the 2^n subsets contain 1 (pair each subset containing 1 with the same subset with 1 removed). So

$$\mathrm{P}(1 \in A_1) = \tfrac{1}{2}.$$

Part (ii)

For an integer i, let E_i be the event that i is *not* in both A_1 and A_2. Then $A_1 \cap A_2 = \varnothing$ exactly when every E_i happens. By independence,

$$\mathrm{P}(i \in A_1 \text{ and } i \in A_2) = \tfrac{1}{2} \cdot \tfrac{1}{2} = \tfrac{1}{4}, \qquad \text{so} \qquad \mathrm{P}(E_i) = \tfrac{3}{4}.$$

The event E_i depends only on the coins for integer i, so $E_1, \ldots, E_n$ are independent, and

$$\mathrm{P}(A_1 \cap A_2 = \varnothing) = \mathrm{P}(E_1) \cdots \mathrm{P}(E_n) = \left(\tfrac{3}{4}\right)^n,$$

as required.

The same argument works for any number of sets. An integer i lies in $A_1 \cap A_2 \cap A_3$ with probability $(\frac{1}{2})^3 = \frac{1}{8}$, so it avoids the intersection with probability $\frac{7}{8}$, and

$$\mathrm{P}(A_1 \cap A_2 \cap A_3 = \varnothing) = \left(\tfrac{7}{8}\right)^n.$$

In general i lies in all m sets with probability 2^{-m}, so

$$\mathrm{P}(A_1 \cap A_2 \cap \cdots \cap A_m = \varnothing) = \left(1 - \frac{1}{2^m}\right)^n.$$

Part (iii)

Now $A_1 \subseteq A_2$ means that no integer is in A_1 but missing from A_2. For each integer i there are four equally likely possibilities for (in A_1?, in A_2?), and only one of them, "in A_1, not in A_2", is forbidden. So each integer behaves well with probability $\frac{3}{4}$, independently, and

$$\mathrm{P}(A_1 \subseteq A_2) = \left(\tfrac{3}{4}\right)^n.$$

(This matches part (ii): $A_1 \subseteq A_2$ exactly when A_1 and the complement of A_2 are disjoint, and the complement of a uniform random subset is again uniform.)

For $A_1 \subseteq A_2 \subseteq A_3$, look at the membership pattern of one integer i in (A_1, A_2, A_3), writing 1 for "in" and 0 for "out". The chain condition says that once i is in some A_k it is in all the later ones too, so the pattern must be one of

$$000, \quad 001, \quad 011, \quad 111.$$

That is 4 of the 8 equally likely patterns, so each integer is fine with probability $\frac{4}{8} = \frac{1}{2}$, and

$$\mathrm{P}(A_1 \subseteq A_2 \subseteq A_3) = \left(\tfrac{1}{2}\right)^n.$$

For a chain of m sets, an allowed pattern is a block of 0s followed by a block of 1s: it is decided by the position where the 1s start, which is one of $1, 2, \ldots, m$, or "never". That gives $m+1$ allowed patterns out of 2^m, so

$$\mathrm{P}(A_1 \subseteq A_2 \subseteq \cdots \subseteq A_m) = \left(\frac{m+1}{2^m}\right)^n.$$

Checks

- With $m = 2$ the general formulas give $(1 - \frac{1}{4})^n$ and $(\frac{3}{4})^n$, and with $m = 3$ they give $(\frac{7}{8})^n$ and $(\frac{4}{8})^n$, matching the special cases. With $m = 1$ they give $(\frac{1}{2})^n$ (only $A_1 = \varnothing$ works) and 1 (one set is always a chain), as they should.
- With $n = 1$ everything can be listed by hand. For example, $A_1 = A_2 = \{1\}$ is the only one of the four choices with a non-empty intersection, giving $\frac{3}{4}$.
- A computer check listing every choice of sets confirmed all the formulas for small n and m. For example, with $n = 3$, $\mathrm{P}(A_1 \cap A_2 = \varnothing) = \frac{27}{64}$, and $\mathrm{P}(A_1 \subseteq A_2 \subseteq A_3) = \frac{1}{8}$.

Common slips

- Answering $\left(\frac{1}{2}\right)^n$ for $\mathrm{P}(A_1 \subseteq A_2)$ by arguing that each element of A_1 must lie in A_2, without allowing for the size of A_1 itself being random.
- Counting the chain patterns as m rather than $m+1$, by forgetting the pattern in which the integer is in none of the sets.

Beyond STEP

Representing a random subset as a string of independent coin tosses (its *indicator vector*) is a standard tool in combinatorics, and the chain count here is the start of the theory of *posets* and the Boolean lattice. The same ideas give the expected number of chains, or the probability that random sets cover S, with very little extra work. Grimmett and Welsh, *Probability: An Introduction*, develops independence and indicator variables in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.